From Interpretable to Black-Box: Dynamic Model Identification for Small USV Control Design

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***Abstract*—Effective low-level feedback control of small under-actuated uncrewed surface vehicles (USVs), specifically vessels under seven meters with a single thruster and rudder, requires dynamic models that provide both actionable design insight and predictive accuracy. Tow-tank testing is impractical at this scale, so parameters must be identified from limited field trials. Existing system identification methods span a wide range of model complexity but offer little guidance on how much fidelity is enough. Sufficiency is only meaningful relative to a stated purpose. Absent that specification the default is a bias toward more fidelity. We propose a methodology built around a deliberate spectrum of three model levels evaluated on common experimental data: first-order linear single-input single-output transfer functions; a physics-based nonlinear state-space formulation with quadratic hydrodynamic drag; and a nonlinear autoregressive with exogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively. Field results demonstrate the approach. First-order models achieve $R^2 = 0.93$ in surge and $R^2 = 0.83$ in yaw rate, and provide sufficient information to approximate a traditional PI+FF compensator design. A margin-based diagnostic makes the sufficiency question computable and returns an interval rather than a threshold: the first-order surge model supports the control design only between 0.35 and 1.28 m/s, under half the tested envelope, bounded below by robust stability and above by loss of closed-loop bandwidth The NARX model reaches one-step prediction $R^2 = 0.998$ in speed and 0.989 in yaw rate on the common trial, yet under recursive simulation on that same trial falls to -1.988 and 0.259, diverging in surge where both structured models remain accurate. Physical structure provides essential regularization at this data scale.**

***Index Terms*—uncrewed surface vehicle, system identification, model spectrum, control-oriented modeling, robustness margin, nonlinear dynamics, underactuated vessels**

I. Introduction

Small uncrewed surface vehicles, here taken as vessels under seven meters driven by a single thruster and rudder, are increasingly deployed for survey, inspection and autonomy research. Low-level speed and heading control is a prerequisite for any higher-level autonomy on such a platform, and that control loop benefits from being designed against a dynamic model.

Which model is the right one depends on what the model is for. The question "how much fidelity is enough" has no answer until "enough for what" has been specified, and because that specification is usually left implicit, practice defaults to a bias toward more fidelity. The reason is asymmetric accounting. The benefit of an added term is easy to measure as an increment in R^2, while its costs are diffuse and rarely tallied.

Those costs are concrete. First, added terms degrade the conditioning of the fit, so parameters that were previously well determined become less certain [1]. That is a loss taken rather than a gain forgone. Second, every added term needs excitation to identify it, so fidelity converts directly into field trials on a platform where trials are the binding constraint. Third, a model with many terms predicts without explaining, and cannot say which term drives the behavior, which is what architecture selection needs from it. Fourth, a high-fidelity model invites the belief that computed gains will transfer to the vehicle, which can make a team less likely to plan field validation. That last one is a schedule and safety risk rather than a modeling one.

Broadly this is the overfitting problem, and driving R^2 upward is a naive objective once it is separated from what the model is for. Machine learning made the trade visible by convention through held-out data and pays it down through regularization [1]. Held-out scoring is nonetheless a weaker check in this setting, because the parameters and not only the predictions are the deliverable. A maneuvering model can pass a prediction test while its individual coefficients are physically meaningless [2]. That motivates diagnostics which compare across model structures rather than scoring one structure against held-out data.

Prior work is not naive about this, and the most careful example shows where the difficulty sits. Sonnenburg and Woolsey [3] build genuine quantitative machinery for sufficiency, with a cost function evaluated on held-out validation maneuvers and tabulated per model and per regime. The final decision is nonetheless made by judgment, discarding the model that scores best because its two extra parameters are not worth the slight edge in turn-rate estimation. This is not a lapse on their part. Their cost function scores the benefit of added parameters, and no comparable metric exists for the cost, so there is nothing for the judgment to be replaced by. The quantitative apparatus stops one step short of the decision it was built to support.

Two distinct claims bear on this, and the second is the one we need. Box's claim is epistemic: correctness is unattainable, so a wrong model cannot be made correct by excessive elaboration, and overparameterization is often the mark of mediocrity [4], [5]. Parsimony follows from the impossibility of truth. The stronger claim is about decisions. Additional correctness is often genuinely available and should still be declined when it trades against the outcome. Skelton put this in control terms in 1989: model errors need not be small but simply appropriate for the control design [6]. Fidelity is not the criterion, fitness for the design decision is. Skelton's criterion has been available for thirty-five years and remains qualitative, since "appropriate" is not a computable predicate. The identification-for-control tradition [7], [8] formalized the idea for experiment design without producing a field test that a small-boat practitioner can run.

What a model of this class can support, in rough order of increasing demand, is a ladder of four design decisions. D1, choose a control architecture, is the weakest requirement: the model need only get the dominant structure right, meaning how many states, which couplings matter and where the dominant timescale sits. D2, verify the architecture end to end, catches structural and integration faults on the bench rather than on the water. D3, compare architectures relatively, asks what a feedforward term or a rate filter is worth, and differences between architectures survive model error far better than absolute values do. D4, rehearse the tuning process, builds the procedure and tooling that will be used to tune in the field, where the deliverable is the workflow rather than the gains it produces. What a model of this class cannot support, and we label this D5, is identifying parameters and designing the controller in simulation and then expecting zero-shot transfer. Saying this plainly is unusual and it is the honest answer to why one should model at all when re-tuning happens in the field anyway. The purchase is D1 through D4. It is not the gains.

Our scope is the single thruster plus rudder vessel under seven meters with field-identified parameters. Underactuated rudder steering is the more interesting case, because control authority is an explicit function of forward speed, so the steering and speed channels cannot be treated as independent design problems. Differential or holonomic thrust reframes the problem rather than simplifying it and removes precisely this coupling. It is also the more representative case for the installed base of small craft, even though the USV literature is dominated by differential-thrust research platforms. It is distinct as well from large vessels, where scale-model tow-tank testing is a reasonable route to parameters and identification is not the binding constraint.

We fix the target control architecture as deliberately as we fix the vehicle class. The models here exist to support the low-level stabilization loops of a production autopilot, specifically the speed and yaw-rate compensators of ArduPilot, which are proportional-integral loops with a feedforward term. That choice is not incidental to the argument. Sufficiency is only answerable once the compensator structure is given, because the question then becomes concrete: does a change of model level move the gains, the achievable bandwidth or the tuning procedure of that specific compensator? A model that alters none of the three is inert for this purpose however much better it fits. Every diagnostic in Sec. III, and Diagnostic 4 in particular, is posed against that fixed architecture.

This paper contributes a three-level model spectrum methodology for this vehicle class; cross-model diagnostics that identify structurally salient dynamics; a margin-based sufficiency test that reports the operating range over which added model fidelity stops changing the control design, replacing a judgment call with a computed interval; an empirical demonstration of when physical structure is required, by way of black-box simulation failure under limited data; and a targeted data-collection framework motivated by the diagnostic gaps.

II. Closely Related Work

A. Response Models as the Minimal Description

The minimal control-oriented description of a steered vessel is Nomoto's first-order steering model [9], which reduces the steering response to a gain and a time constant and remains the standard starting point in the modern formulation [10]. The identification tradition built on it was founded by Åström and Källström [11], who estimated steering dynamics from full-scale trials rather than from tank data. That lineage is the direct ancestor of our Level 1.

B. Physics-Based Maneuvering Models and Identifiability

Physics-based maneuvering models in the Abkowitz tradition [12] expand the hydrodynamic forces in a polynomial series whose coefficients carry physical meaning. The difficulty is that those coefficients resist identification from field data. Multicollinearity among regressors makes individual coefficients unstable even when the aggregate prediction is good [1], [2], [13]. This is the identifiability barrier that makes "add another term" a more expensive proposition than its ΔR^2 suggests.

C. Control-Oriented Identification of Small Craft

The nearest antecedent to this work is Sonnenburg and Woolsey [3], [14], who identify and compare a family of models for a small planing craft and evaluate them for control design. Related efforts span autonomous surface craft and small workboats, including [15]–[21]. The common pattern is a comparison across candidate structures followed by selection of one, after which the discarded models stop contributing.

D. Data-Driven and Grey-Box Alternatives

Data-driven and grey-box marine models have grown quickly, spanning recurrent architectures [22], physics-informed networks [23] and gradient-boosted ensembles [24]. The NARX formulation we adopt at Level 3 descends from the neural identification literature of the early 1990s [25] and its delayed-input variants [26]. Evaluation practice is uneven across this literature. The distinction relevant to control is between one-step prediction, which reads measured state history at every sample, and recursive simulation, which feeds predicted states back and is what a controller design loop requires. Woo *et al.* [22] do validate through forward simulation on held-out maneuver records, so their work is not an instance of the gap, and we report both metrics for the same reason.

E. Positioning

Three gaps motivate this work. Prior work resolves sufficiency by selecting one model rather than retaining the set. Evaluation of data-driven marine models is inconsistent about the one-step versus recursive distinction. The small rudder-steered vessel is underrepresented relative to differential-thrust research platforms.

III. Model Spectrum Methodology

The three levels are not a fidelity ladder to be climbed and discarded. They map onto the decision ladder of Sec. I: Level 1 serves D1, Level 2 serves D2 and D3, and Level 3 bounds how far D4 could be pushed with more data. The design target throughout is the PI-plus-feedforward stabilization loop identified in Sec. I, so each level is judged by what it contributes to that compensator rather than by goodness of fit.

A. Level 1: First-Order Linear SISO Models

Level 1 treats surge and yaw as decoupled first-order channels,

$$G_v(s) = \frac{K_v}{\tau_v s + 1}, \qquad G_r(s) = \frac{K_r}{\tau_r s + 1}, \tag{1}$$

mapping normalized throttle to forward speed and normalized rudder to yaw rate. Both are identified as discrete ARX(1,1) models by ordinary least squares on the 10 Hz record and converted to continuous time. The parameters are immediately usable: τ sets the achievable loop bandwidth and K sets the loop gain, so a PI controller follows directly.

B. Level 2: Nonlinear State-Space Model

Level 2 retains the decoupled channel structure but replaces linear damping in surge with quadratic hydrodynamic drag,

$$\dot{v} = k_T\, u - d_2\, v|v|, \tag{2}$$

$$\dot{r} = k_\delta\, \delta - d_r\, r, \tag{3}$$

with u the normalized throttle and δ the normalized rudder.

Identification is two-phase, and the phasing is what makes it work. Fitting (2) jointly for k_T and d_2 is poorly conditioned, because thrust and drag are both large and nearly cancelling during steady runs, so the data constrains their difference far better than either term. Instead d_2 is estimated from coasting segments alone, where $u = 0$ and (2) reduces to $\dot{v} = -d_2 v|v|$ with a single unknown. With d_2 fixed, k_T follows from the powered segments by linear least squares.

C. Level 3: Black-Box NARX Model

Level 3 is a fixed-tap residual NARX multilayer perceptron with 19 lagged inputs, one hidden layer of 64 tanh units and two outputs which are state increments rather than absolute states. State history extends 0.5 s into the past, throttle history 2 s and rudder history 1 s. The lag structure is a project-specific implementation choice consistent with delayed-input NARX models [26]. Being multi-input multi-output, the model is free to discover the surge-yaw coupling that Levels 1 and 2 assume away.

Level 3 is a compact black-box reference point in the comparison rather than a competitor, and no claim is made that this architecture is optimal. Its role is to bound the benefit of added complexity at the available data scale. Evaluation reports one-step prediction and recursive simulation separately, following the distinction drawn in Sec. II.

D. Cross-Model Diagnostics

The value of the spectrum is the comparison across levels rather than the three fits individually. Four diagnostics extract that comparison. The first three ask whether a term is present in the data. The fourth asks whether it changes the design.

Diagnostic 1, speed-dependent rudder gain. Replace the constant k_δ in (3) with a speed-scaled $k_\delta v$ and test whether the yaw-rate fit improves.

Diagnostic 2, drag nonlinearity. Fit linear and quadratic damping to the coasting deceleration and compare.

Diagnostic 3, rudder-induced surge drag. Correlate δ^2 against the surge residual of the Level-2 fit.

Diagnostic 4, is the added fidelity inside the robustness margin? The robustness margin of a compensator is a budget for model error, so the test is whether the Level-1 to Level-2 discrepancy fits inside the margin of a controller designed from Level 1 alone. The construction has four steps.

Step 1. Design a nominal compensator from Level 1 only, in the target architecture of Sec. I, so that the test measures a change the deployed autopilot would experience rather than a property of some auxiliary controller. We use an internal model control PI, $C(s) = (\tau_v s + 1)/(K_v \lambda s)$, which cancels the Level-1 pole and gives a nominal loop $L = 1/(\lambda s)$ and complementary sensitivity $T(s) = 1/(\lambda s + 1)$, with λ the intended closed-loop time constant.

Step 2. Linearize Level 2 about operating points spanning the tested envelope. For surge, $\partial(v|v|)/\partial v = 2v_0$ for $v_0 > 0$, so (2) gives a speed-scheduled first-order plant

$$G_2(s, v_0) = \frac{K_2(v_0)}{\tau_2(v_0)s + 1}, \quad \tau_2 = \frac{1}{2d_2 v_0}, \quad K_2 = \frac{k_T}{2d_2 v_0}, \tag{4}$$

in which both the gain and the time constant fall as $1/v_0$.

Step 3. Form the multiplicative model error $\ell(\omega, v_0) = |G_2(j\omega, v_0)/G_1(j\omega) - 1|$ and test robust stability, $\|\ell T\|_\infty < 1$.

Step 4. Report the operating range over which the test passes. The output is not a yes or no but the speeds at which Level 1 stops being sufficient, which is an actionable number for a control designer and a direct answer to "enough for what."

Fig. 1. The experimental platform, a Pro Boat Blackjack 42 monohull of 1.07 m length, afloat during a field trial. Thrust comes from a single propeller and steering from a single rudder, both at the stern. The autopilot, the GNSS antenna and the radio links are mounted above the deck on the aft half of the hull.

Step 3 alone turns out to be insufficient, and Sec. V reports why. Robust stability is a statement about whether the loop remains stable, and quadratic drag is stabilizing, so at high speed the Level-1 design becomes sluggish rather than unstable and the stability test keeps passing. We therefore add a performance criterion alongside it: apply the Level-1 compensator to $G_2(s, v_0)$ and compare the achieved closed-loop bandwidth against the design intent $1/\lambda$. The pair of criteria bounds sufficiency from both sides.

IV. Experimental Platform and Data Collection

A. Vehicle

The platform, shown in Fig. 1, is a Pro Boat Blackjack 42, a production radio-controlled monohull of 1.07 m (42 in) length, instrumented to a mass of approximately 5 kg. Propulsion is single-screw: one four-pole water-cooled brushless motor turning a single propeller through a 160 A water-cooled electronic speed controller, with a single rudder driven by one digital servo and an 8S lithium-polymer pack supplying power. The vehicle is therefore underactuated in exactly the sense of Sec. I, with forward thrust on one channel and steering moment on another whose authority depends on forward speed.

Choosing a production hull rather than a purpose-built platform is deliberate. Vessels of this class are what the small-USV installed base looks like, and the identification problem is posed by the vehicle that exists rather than by one designed to be identifiable.

B. Autopilot, Sensing and Logging

Guidance, control and logging run on a Cube Orange autopilot executing ArduRover 4.6.3 (commit `3fc7011a`), with Mission Planner as the ground station. Position and ground speed come from a HERE3 GNSS receiver over DroneCAN, and angular rates from the autopilot's internal inertial measurement unit.

The actuator configuration is confirmed by the parameter set: `SERVO1` is assigned to ground steering and `SERVO3` to throttle, and every other servo output is disabled. There is one thrust input and one steering input, with no differential-thrust path. Commands are logged as `RCOU_C3` (throttle) and `RCOU_C1` (rudder) at 1100–1900 μs PWM and normalized to $[-1, 1]$ about the 1500 μs neutral point. Measured outputs are `GPS_Spd` ground speed at 5 Hz and `IMU_GyrZ` yaw rate at 137 Hz. The trials used here are flown in manual mode, so the logged commands are open-loop operator inputs rather than the output of any autopilot loop.

The autopilot's stabilization layer is the design target named in Sec. I, and it is worth stating concretely because it is what the model levels are ultimately for. Two independent loops run at that layer: throttle to forward speed (`ATC_SPEED_`$\star$) and rudder to yaw rate (`ATC_STR_RAT_`$\star$), each nominally a PID with an added feedforward term. On this vehicle the derivative gain of both loops is set to zero, so the deployed compensator is proportional-integral plus feedforward, matching the structure assumed in Sec. III. The remaining gains are $P = 0.4$, $I = 0.1$, $\mathrm{FF} = 0.2$ on the speed loop and $P = 1.0$, $I = 0.77$, $\mathrm{FF} = 0.2$ on the yaw-rate loop.

The feedforward gain is the point of contact between this paper and the deployed controller. For a first-order channel the feedforward term that produces a commanded steady state is the inverse of the identified DC gain, which is exactly what Level 1 delivers. The deployed surge value of 0.20 sits close to the $1/K_v = 0.16$ implied by the Level-1 fit, which is consistent with the two being the same quantity. The yaw-rate channel does not correspond so directly, since $1/K_r = 1.30$ against a deployed 0.20, and resolving that gap requires the firmware's internal rate scaling rather than anything in the identification. We note the comparison rather than rest any claim on it.

Binary autopilot logs are converted to MATLAB format by a utility that extracts every numeric field of every message type under a `{MSGTYPE}_{field}` naming convention, and all subsequent processing works from that representation.

The configured cruise speed for the vehicle is 5 m/s. The identification trial described below spans 0–2.9 m/s, so it covers roughly the lower half of the speed range the vehicle is intended to operate over. This bears on Diagnostic 4 in Sec. V-D, whose sufficiency interval is narrower still.

C. Data Cleaning

Multi-rate streams are aligned to a common 10 Hz grid by linear interpolation. Records are segmented at mode changes, signal-freshness discontinuities and quality flags, so that no segment spans a transition that the models are not intended to represent.

D. Datasets

Levels 1 and 2 are identified from a single step-response trial, a 130 s working window drawn from a 341 s recording, covering throttle 0–39% and speed 0–2.9 m/s and comprising 1,300 samples at 10 Hz. One 29.3 s gap caused by vessel repositioning is handled by segmentation. This is intentionally minimal, a proof of concept for the workflow rather than a characterization of the vehicle.

Level 3 draws on a larger corpus: 44,133 samples in 70 slow-region segments from 65 source logs. The slow-region rule excludes a complete segment once speed stays above 3.4028 m/s for at least 1 s continuously. Allocation is grouped by source log, so every segment from one recording falls in a single partition and no recording is split across training and validation. Training uses 37 logs, being 40 segments and 28,023 rows. Validation uses every other viable slow-region recording, being 28 logs, 30 segments and 16,110 rows. Ten complete high-speed segments consistent with planing are held aside for later regime-spanning work and are not used in the results below. That label is empirical GPS speed over ground rather than a verified physical planing transition.

For direct comparison across levels, the frozen Level 3 model is also evaluated on the same 130 s trial used for Levels 1 and 2. Its 20-sample lag history leaves 1,279 scored rows against the 1,300 rows available to Levels 1 and 2.

V. Results

A. Level 1

Ordinary least squares on the step-response trial gives

$$G_v(s) = \frac{6.42}{1.95s+1}, \qquad G_r(s) = \frac{0.771}{0.471s+1}, \qquad (5)$$

with simulation $R^2 = 0.930$ in surge and 0.826 in yaw rate. The surge time constant of 1.95 s and the yaw time constant of 0.471 s differ by a factor of four, which alone settles the architecture question at D1: the two channels can be given separate loops at separate bandwidths.

B. Level 2

Two-phase identification gives

$$\dot{v} = 3.963\, u - 0.438\, v|v|, \qquad (6)$$

$$\dot{r} = 1.474\, \delta - 1.912\, r, \qquad (7)$$

with $R^2 = 0.909$ in surge and 0.820 in yaw rate. Both are marginally below Level 1 on this trial, which is expected: Level 1 is fit to minimize exactly this error on exactly this data, while Level 2 estimates its drag coefficient from coasting segments alone. The useful comparison is what the structure exposes rather than the R^2, and Sec. V-D takes that up.

Estimating d_2 from coasting is what makes the fit usable. The joint fit is poorly conditioned for the reason given in Sec. III, and the resulting parameters are sensitive to the powered-segment weighting in a way the coasting estimate is not.

C. Level 3

On the full slow-region validation set of 30 segments from 28 held-out logs, the one-step macro-NRMSE is 0.084 and the recursive macro-NRMSE is 0.425. Speed R^2 falls from 0.998 one-step to 0.836 recursive, and yaw-rate R^2 from 0.991 to 0.768. Recursive simulation covers every validation sample. Fig. 3 shows the surge channel across that set.

The comparison on the common trial is sharper. Level 3 reaches one-step R^2 of 0.998 in speed and 0.989 in yaw

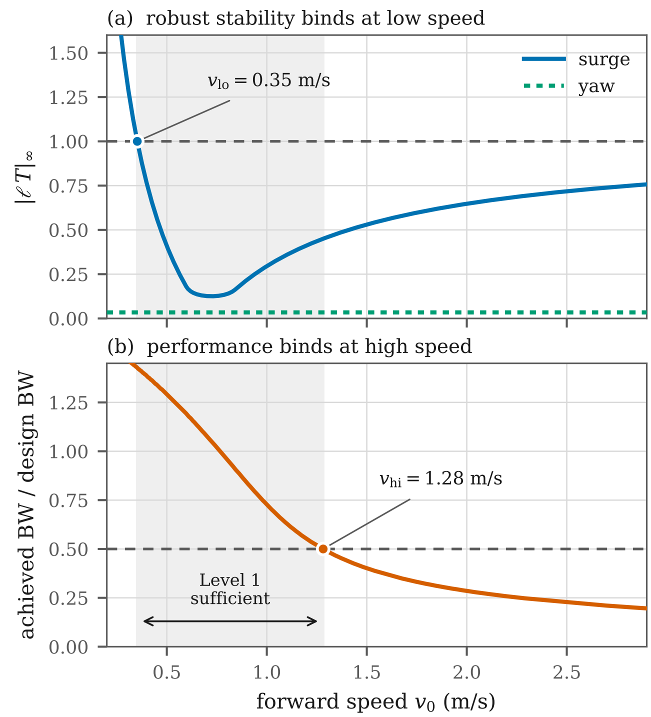

Fig. 2. Diagnostic 4. (a) Robust stability of the Level-1 compensator against the Level-2 linearization binds at low speed, where the quadratic-drag linearization is singular. (b) Achieved closed-loop bandwidth binds at high speed. The shaded band is the interval over which the Level-1 model supports the design. The yaw channel (a, dotted) never approaches the limit, so Level 2 changes nothing for yaw control.

rate, above both structured models on the same data. Under recursive simulation on that same trial it falls to −1.988 in speed and 0.259 in yaw rate. A negative R^2 means the model is a worse predictor than the mean of the measured signal. Fig. 4 shows what that looks like: the surge trajectory drifts below zero and stays there for tens of seconds while Levels 1 and 2 track the measured response.

The gap between one-step and recursive is the result. One-step prediction reads the measured state at every sample, so a model can score 0.998 while having learned little more than that the state changes slowly. Recursive simulation removes that support, and the small per-step increment errors accumulate without any physical constraint to arrest them. Levels 1 and 2 cannot drift this way because the drag term bounds the surge state structurally.

D. Cross-Model Diagnostic Summary

Diagnostic 1 finds no improvement from a speed-dependent rudder gain ($\Delta R^2 = -0.03$). The trial holds forward speed nearly constant during the rudder steps, so the regressor is barely excited. This flags a gap in data coverage rather than absence of the effect, and it is the primary target of the next collection campaign.

Diagnostic 2 finds quadratic damping clearly dominant, with $\Delta R^2 = +0.20$ over the linear fit, $d_1 \approx 0$ and $d_2 = 0.438\ \mathrm{m}^{-1}$.

This is the direct empirical warrant for the Level-2 structure.

Diagnostic 3 finds rudder-induced surge drag negligible, with a correlation of $r = -0.08$ between δ^2 and the surge residual. The coupling term is not warranted by this data.

Diagnostic 4 is reported in Fig. 2, computed with $\lambda = 1.0$ s. The result is an interval rather than a threshold. Robust stability fails below $v_0 = 0.35$ m/s, and this lower bound has the closed form

$$v_{\text{lo}} = \frac{k_T}{4d_2K_v}, \tag{8}$$

which is independent of λ because $|T| \leq 1$ places the supremum of $\ell|T|$ at DC. The mechanism is the $1/v_0$ singularity in (4): quadratic drag provides no linear damping at zero speed, so the linearized gain and time constant both diverge and no first-order fit can track them.

At the upper end the stability test never fails. At 2.9 m/s the model error is still within budget at $\|\ell T\|_\infty = 0.76$. What fails is performance: the Level-1 compensator applied to the Level-2 plant retains only 20% of its intended bandwidth at 2.9 m/s, crossing 50% at $v_{\text{hi}} = 1.28$ m/s. Quadratic drag is stabilizing, so the Level-1 design at speed is conservative rather than unsafe. The failure mode is sluggishness.

Taken together, the Level-1 surge model supports the control design only over roughly $[0.35, 1.28]$ m/s, under half of the tested envelope. Both bounds are insensitive to the design knob: sweeping λ from 0.5 to 3 s moves v_{hi} only between 1.23 and 1.61 m/s and leaves v_{lo} unchanged.

Two by-products of the construction are worth recording. The Level-1 fit is anchored where it coincides with the Level-2 linearization, at $v_0 = 0.70$ m/s for the gain and 0.59 m/s for the time constant, which is well below the top of the envelope and explains why the single-trial linear fit degrades at speed. And the high-frequency limit of the model error, $k_T\tau_v/K_v = 1.20$, is independent of v_0, so the discrepancy at high frequency is a fixed property of the two fits rather than an operating-point effect.

For yaw the picture is different. Level 2 linearizes to $r/\delta = 0.771/(0.523s + 1)$ against Level 1's $0.771/(0.471s + 1)$: identical DC gain and an 11% difference in time constant, giving $\|\ell T\|_\infty = 0.034$. Level 2 buys nothing for yaw control across the whole envelope, and we say so rather than let the added structure imply otherwise.

VI. Discussion

A. What the Spectrum Reveals

For this vehicle class and at the speeds tested, surge and yaw are weakly coupled, so decoupled SISO controllers are a defensible starting point. Quadratic drag is significant, which means a speed-hold controller needs nonlinear feedforward and a purely linear controller will show speed-dependent steady-state error. Whether the rudder gain scales with forward speed can be neither confirmed nor denied from this data, and that is the single largest gap the expanded dataset must fill.

No one of the three models exposes all of this. Level 1 alone gives no reason to suspect the drag is quadratic. Level 2 alone does not reveal that its extra structure is inert for yaw. Level 3 alone would report an excellent one-step score and mislead anyone who took it to simulation.

B. Implications for Compensator Architecture

Level 1 supports a baseline PI structure with directly identified gains, valid over the interval Diagnostic 4 returns. Level 2 justifies nonlinear feedforward on the surge channel, and Diagnostic 4 says at what speed it becomes necessary. On the yaw channel Level 2 changes nothing, so the added structure should not be carried into the design. Gain scheduling on the rudder loop remains open and is gated on Diagnostic 1 rather than on Level 2. Level 3's role here is to bound what is available beyond the fixed architecture. A black-box model with full freedom to discover coupling that the structured levels assume away is the natural precursor to model-predictive or learning-based control, and its recursive failure on the common trial says plainly that the data does not yet support moving past proportional-integral plus feedforward. That is a useful negative result for a practitioner deciding where to spend effort: at this data scale the return lies in better excitation for the structured levels rather than in a richer controller.

C. Predictive Fit Against Simulation Usefulness

The Level-3 result is the paper's sharpest empirical claim. A model can be the most accurate one-step predictor in the comparison and simultaneously the least useful simulator, and the two properties are measured by numbers that differ by more than two units of R^2 on identical data. Any evaluation that reports only one-step accuracy is silent about the property a control designer needs. Physical structure is useful here because it constrains what the model can do once the measurements are taken away. Fitting better is not what it contributes.

D. Returning to the Decision Ladder

None of the three levels reaches D5, and saying so plainly is the honest position. Even the highest-fidelity model here cannot eliminate field tuning. The defensible claim is D1 through D4: the spectrum says which architecture to build, permits end-to-end verification, supports relative comparison between candidates and identifies where in the parameter space field tuning will be needed, so that the tuning workflow can be built in advance.

E. Limitations

Levels 1 and 2 rest on a single trial covering 0–2.9 m/s and throttle to 39%, so the identified parameters are not a characterization of the vehicle across its envelope. Diagnostic 4 inherits that limitation, since the Level-2 linearization it uses is only as good as the parameters behind it. The common-trial comparison in Sec. V-C rests on one 130 s record, which is enough to demonstrate the divergence but not to quantify how often it occurs. Level 3's training corpus is broader but restricted to the slow region by construction, so nothing here speaks to behavior near or above the planing transition. Since

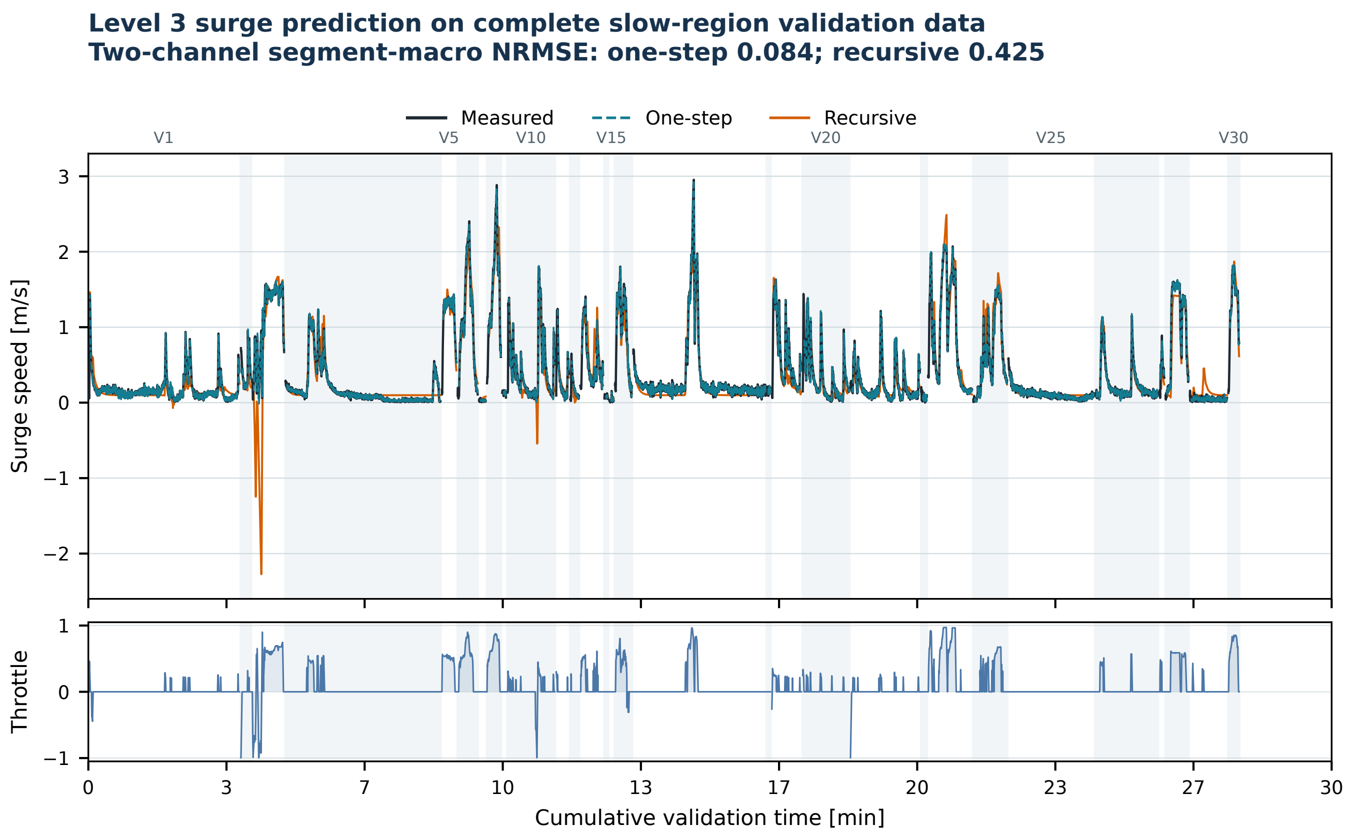

Fig. 3. Level 3 surge prediction on the complete slow-region validation set of 30 segments from 28 held-out source logs. Each segment is initialized with 2 s of measured history. One-step prediction continues to use measured state history; recursive simulation feeds predicted states back while replaying recorded throttle. Alternating backgrounds and short gaps mark segment boundaries. Reported NRMSE values are segment-macro averages over speed and yaw-rate errors normalized by training-only channel scales.

the configured cruise speed is 5 m/s, none of the three levels is characterized over the upper half of the vehicle's intended operating range. All results come from a single vehicle, so generalization across hull form and scale within this class is untested here.

VII. Conclusion

We have presented a three-level model spectrum methodology for small underactuated USVs, identified from field trials and evaluated on common data. The central finding is that the spectrum collectively provides more control design guidance than any single model. Simple models are a persistent design tool rather than a stepping stone to be discarded once a better fit is available.

The margin-based sufficiency test converts Skelton's qualitative criterion into a computed interval for this vehicle class. For the surge channel it returns $[0.35, 1.28]$ m/s, bounded below by robust stability and above by loss of closed-loop bandwidth, and it reports that the same test is satisfied everywhere on the yaw channel, where the added Level-2 structure is inert.

Future work runs in three directions. The first is data: multiple trials spanning the full throttle range and dedicated rudder sweeps at fixed speeds, which is what resolves the speed-dependent rudder gain that Diagnostic 1 could not settle. The second is a stronger basis for cross-level comparison. The present common trial is a single 130 s record, and a larger multi-trial common evaluation set would let the Level-3 divergence be quantified rather than demonstrated. Related to this, the sensitivity of each level to the quantity of training data is unstudied here and deserves direct treatment, particularly across displacement, semi-planing and planing regimes, where the amount of data needed to identify a usable model may differ substantially by regime. The third is to treat the robustness margin as a formal sufficiency criterion across the whole spectrum, computing for each design decision the model level at which further fidelity is provably inert. That would replace the expert judgment identified in Sec. II with a computable bound and it generalizes past this vehicle class.

The dataset and analysis notebooks will be released.

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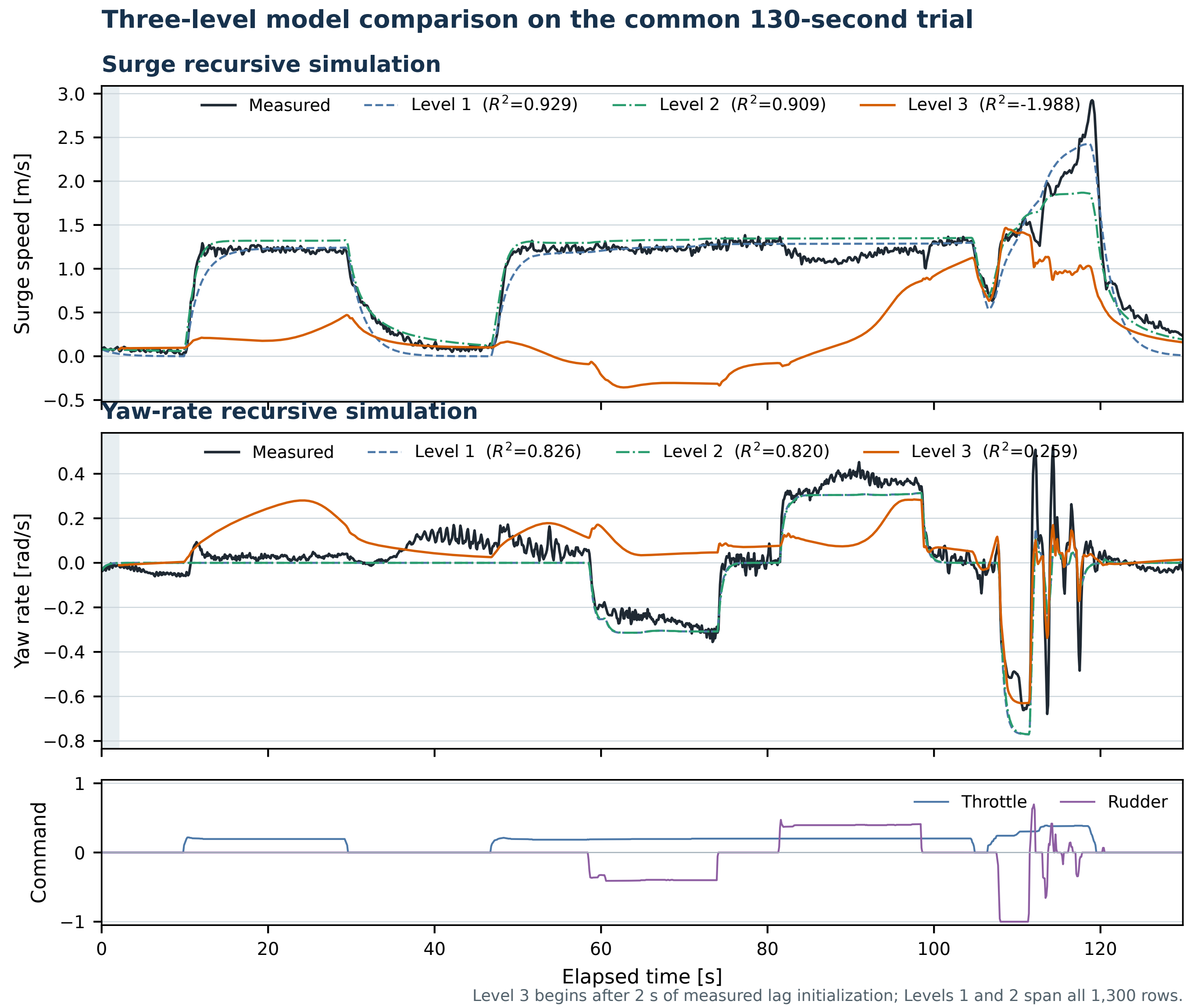

Fig. 4. All three levels under recursive simulation on the common 130 s trial. Levels 1 and 2 track the measured response; Level 3 diverges in surge, driving the predicted speed below zero for tens of seconds despite reaching the highest one-step accuracy of the three. Level 3 begins after 2 s of measured lag initialization, so it scores 1,279 rows against 1,300 for Levels 1 and 2.

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